

# An Analysis of osu! Beatmap Popularity: Uncovering Patterns of Player Engagement

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**Abstract**—Digital content is a powerful tool for analyzing user engagement and digital content popularity. This study aims to uncover the data-driven formulas behind successful rhythm game design. The dataset includes 61,686 beatmaps created, played, and rated by millions of users. Exploratory Data Analysis (EDA), unsupervised clustering, and association rule mining were employed to explore the distribution of key features within the dataset. The data revealed that popularity is driven by specific tempos and difficulties, and that player intent bifurcates into distinct “grind” and “experience” categories. These findings have direct implications for community content creators and rhythm game developers. Rather than trying to design a map that does everything, creators can use these blueprints to target specific audiences, designing mechanically dense maps for retention, or accessible, aesthetic maps for appreciation.

## I. INTRODUCTION

The rhythm game osu! boasts a massive, community-driven catalog of playable levels known as “beatmaps.” With hundreds of thousands of maps created, played, and rated by millions of users, the ecosystem provides a rich dataset for analyzing user engagement and digital content popularity. However, popularity in gaming is often treated as a monolith.

This project explores a dataset of osu! beatmap statistics from 2007 to August 2019 (sourced from Github) [1] to uncover the data-driven formulas behind successful rhythm game design. The dataset originally contained 83,641 rows and 37 attributes characterizing map metadata, difficulty, and player engagement metrics (e.g., playcount, passcount, favourites).

The analysis is guided by two primary discovery questions:

*A. What map and song characteristics are most commonly shared among the most popular beatmaps?*

Instead of assuming “harder = better,” this explores how parameters like BPM, total beatmap length, and difficulty ratings mathematically drive player attention.

*B. Are there distinct groups of songs that achieve popularity in different ways?*

This explores the hypothesis that engagement is multi-dimensional, separating maps into categories such as “grinds” (high replay) versus “experiences” (high appreciation).

## II. METHODS

To answer these questions, a combination of Exploratory Data Analysis (EDA), unsupervised clustering, and association rule mining was employed.

### A. Preprocessing and Data Cleaning

To ensure data quality, non-standard game modes (modes other than 0), pending, work-in-progress, and graveyarded (abandoned) maps were removed, leaving only ranked, loved, or qualified maps. Outliers resulting from inaccurate data collection ( $\text{BPM} \geq 500$ ,  $\text{difficulty} \geq 11$  stars) were dropped. To filter out low-effort or anomalous maps, song length boundaries were established. Maps shorter than 40 seconds or longer than two standard deviations above the mean were removed. Furthermore, a new beatmap attribute `drain_ratio` (hit length divided by total length) was calculated. Maps with a drain ratio below 0.60 were removed, ensuring the dataset only contained maps with at least 60% active gameplay, filtering out tracks with excessively long intros, pauses, and breaks. This resulted in a refined dataset of 61,686 high-quality beatmaps.

### B. Exploratory Data Analysis

EDA was conducted to understand the distribution of key variables. Because engagement metrics like playcount and favourite\_count follow extreme exponential distributions (a few mega-hits garnering millions of plays), a logarithmic transformation ( $\log(x + 1)$ ) was applied to normalize the data for visualization and algorithmic processing. A custom metric, `replay_ratio`, was calculated by dividing playcount by passcount (plus 1 to avoid division by zero). This metric isolates retry behavior, serving as a proxy for how heavily a map is “grinded.”

### C. Mining Techniques

*1) K-Means Clustering:* To discover natural groupings within the maps, K-Means clustering was applied to five key scaled features: log-transformed replay ratio, log-transformed favourite count, difficulty rating, BPM, and total length. While the Silhouette score suggested an optimal  $k = 2$ , this resulted in an overly broad generalization of the data. By analyzing the Elbow Method (WCSS) and utilizing domain context,  $k = 4$  was selected to capture practical, nuanced differences between map types. DBSCAN was also explored but abandoned; due to the dense, continuous nature of rhythm game metadata, DBSCAN grouped nearly 55,000 maps into a single cluster, failing to provide actionable insights.

2) *Association Rule Mining (Apriori)*: To find specific “successful combinations” of traits, the Apriori algorithm was implemented. Continuous variables were discretized into categorical bins based on domain standards: Length (Short, Standard, Marathon), BPM (Slow, Standard, Fast), and Difficulty (Easy/Normal, Hard/Insane, Expert). The engagement metrics (replay\_ratio and favourite\_count) were discretized into “High” (top 25th percentile) and “Normal” (bottom 75th percentile). A minimum support of 0.04 and a confidence threshold of 0.5 were used to extract high-lift rules.

### III. RESULTS

#### A. EDA Findings

Initial visualizations revealed strict preferences in the player base. Lower BPM maps (75–150) are significantly under-represented in the top 10% of most-played maps, whereas a massive spike in popularity occurs at the 175 BPM range. Furthermore, a scatterplot of difficulty versus playcount highlighted a distinct “sweet spot”: maps between 4 and 6 stars attract the largest player base, balancing accessibility with mechanical challenge. Plotting the Replay Ratio against

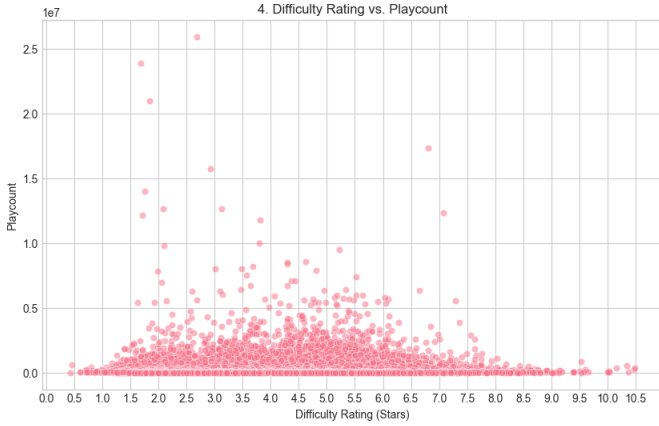


Fig. 1. Difficulty vs Playcount

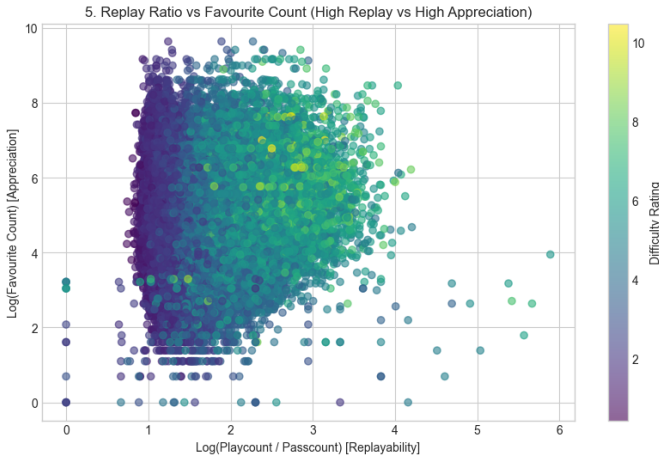


Fig. 2. Replay Ratio vs Favourite Count

Favourite Count visually confirmed the existence of diverging map types. Maps clustered in the bottom-right represented endless retries (high difficulty), while the top-left represented high favorability without repetitive grinding.

#### B. K-Means Clustering Results

The K-Means algorithm ( $k = 4$ ) successfully segmented the maps into distinct behavioral profiles:

- Cluster 2 (“The Grind”): Characterized by the highest replay ratio (2.25), highest difficulty ( $\sim 5.04$  stars), and high BPM ( $\sim 176$ ). These are the mechanically demanding maps that get tried over and over again.
- Cluster 0 (“The Experience”): Characterized by a very low replay ratio (1.27), lower difficulty ( $\sim 2.42$  stars), but the absolute highest average favourite count. These are maps where something other than difficulty attracts players.

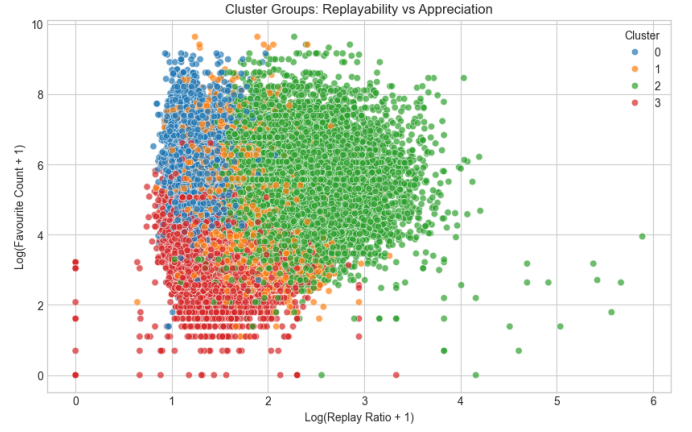


Fig. 3. K-Means Clustering Results

#### C. Apriori Algorithm Results

The Apriori algorithm generated highly robust statistical associations that corroborated the clustering results.

- Rule 3 demonstrated that if a map is categorized as Expert ( $>5.5$  stars), there is a 90% confidence it will result in High Replay, yielding a lift of 3.60.
- Rule 2 showed that combining Expert difficulty with Standard BPM (150-200) increases this confidence to 91% (Lift: 3.63). Crucially, across the top 15 highest-lift rules generated by the algorithm, High Replay was universally paired with Normal Appreciation. There was not a single high-lift rule that successfully linked High Replay with High Appreciation.

| antecedents                                                                       | consequents                          | support                                                      | confidence | lift      |
|-----------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------|------------|-----------|
| low_category_Standard (150-200)                                                   | difficulty_category_Expert (5.5-5.7) | low_category_Standard (150-200), replay_category_High Replay | 0.84       | 0.12 1.96 |
| difficulty_category_Standard (1.5-1.7)                                            | replay_category_High Replay          | difficulty_category_Expert (5.5-5.7)                         | 0.84       | 0.12 1.96 |
| difficulty_category_Standard (1.5-1.7), appreciation_category_Normal Appreciation | replay_category_High Replay          | low_category_Standard (150-200), replay_category_High Replay | 0.84       | 0.12 1.96 |
| low_category_Slow (100-120), replay_category_High Replay                          | difficulty_category_Expert (5.5-5.7) | low_category_Standard (150-200), replay_category_High Replay | 0.84       | 0.12 1.96 |
| low_category_Slow (100-120), replay_category_High Replay                          | difficulty_category_Expert (5.5-5.7) | low_category_Standard (150-200), replay_category_High Replay | 0.84       | 0.12 1.96 |
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Fig. 4. Top 15 Apriori Rules.

#### IV. INTERPRETATION

The findings provide clear, data-backed answers to the initial discovery questions.

Regarding the characteristics of popular beatmaps, popularity is not random; it adheres to a strict mechanical blueprint. The combination of 150–200 BPM and  $>3.5$  star difficulty is the undisputed driver of raw playcount. This represents the standard rhythm game tempo and the mechanical threshold where players begin actively practicing to improve their skills.

Regarding the distinct paths to popularity, the algorithms mathematically prove that *osu!* beatmaps achieve success through mutually exclusive avenues. K-Means isolated “The Grind” (Cluster 2) from “The Experience” (Cluster 0), and Apriori confirmed this by showing a complete lack of association between heavy replayability and high appreciation. Players repeatedly play Hard/Insane maps to achieve mechanical mastery or farm performance points (PP); they do not necessarily favourite them. Conversely, highly favoured maps are played for their artistry, storyboards, or musicality, entirely outside of the high-difficulty retry loop. Popularity in digital content is therefore deeply contextual to player intent.

#### V. CRITICAL ASSESSMENT

##### A. Validity and Robustness

The findings demonstrate strong internal validity because two distinct unsupervised learning techniques (Clustering and Association Rules) independently corroborated the same conclusion regarding the divergence of replayability and appreciation. However, the decision to partition K-Means to  $k = 4$  despite the Silhouette score recommending  $k = 2$  introduces a degree of subjective bias. While  $k = 4$  yielded more actionable domain insights, it acknowledges that the boundaries between these map categories are somewhat porous rather than rigidly separated. The failure of DBSCAN further validates that beatmap statistics exist on a dense continuum rather than in perfectly isolated spatial islands.

##### B. Limitations

Several limitations bound this analysis. First, the dataset concludes in August 2019. The *osu!* meta, mapping styles, and player skill ceilings have evolved significantly since then; the “sweet spot” of 4–6 stars may have shifted higher in recent years. Second, the dataset lacks qualitative attributes. We can mathematically identify an “Experience” map, but we lack the data (e.g., storyboard file sizes, user reviews, visual spacing metrics) to analyze why it is aesthetically appreciated. Finally, playcount is a global aggregate. It cannot distinguish between 10,000 unique players playing a map once, and 100 dedicated players grinding a map 100 times, which blurs the exact nature of the engagement.

##### C. Ethical Considerations

While analyzing video game statistics carries low immediate risk to human welfare, there are ethical considerations regarding how these insights are applied. Uncovering the exact mathematical blueprint for “High Replay” engagement (175

BPM, 5+ stars) could encourage “meta-chasing.” If creators algorithmically optimize their content solely for maximum retention and plays, it could stifle creative diversity, filling the ecosystem with homogeneous content. Furthermore, metrics optimized for endless retries rely heavily on player frustration and repetitive failure, inadvertently promoting addictive, grind-heavy behaviors that can lead to physical strain or unhealthy gaming habits.

#### VI. CONCLUSION

This analysis successfully mapped the landscape of user engagement within *osu!* beatmaps. By pushing past basic frequency metrics, the data revealed that popularity is driven by specific tempos and difficulties, and that player intent bifurcates into distinct “grind” and “experience” categories.

These findings have direct implications for community content creators and rhythm game developers. Rather than trying to design a map that does everything, creators can use these blueprints to target specific audiences, designing mechanically dense maps for retention, or accessible, aesthetic maps for appreciation. Future work should involve integrating post-2019 data to analyze how these engagement blueprints shift over time, and incorporating individual player histories to better separate unique plays from repetitive grinding.

#### REFERENCES

- [1] <https://github.com/jxu/osu-stats/releases/tag/2019-08-01>